# AGI ALPHA FLYWHEEL: The Enterprise That Invents Its Next Advantage

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## Abstract

We connect versioned offerings and bounded enterprise decisions to an existing reasoning/tool executor, optional workflow-learning Gym, protected capability service and disjoint cost ledger. An internal engineering artifact is executed and independently checked. The frozen workflow is tested on 12 fresh finite problems. The system preserves the economically defensible negative result: admission is rejected, additional investment is declined, and an adjacent offering remains a hypothesis. A separate fictional-authority episode tests actual protected loading and invocation. Neither episode establishes live frontier-model advantage, actual customer demand, complete-cost profit or sustained compounding.

## Observed milestone record

| Milestone | Recorded outcome |
|---|---|
| Internal useful delivery | Executed; finite independent acceptance |
| Optional Gym | Completed; executable v1 to v2 change |
| Fresh technical transfer | 12/12 accepted; less search work on 11 |
| Logical search-work totals | v1: 14,022; v2: 3,352 |
| Net Alpha admission | Rejected; competent frontier Beta unmeasured |
| Commercial authorization | Withheld; demand unmeasured |
| Next decision / adjacent offering | Hold funds / hypothesis |
| Actual provider calls / collections | 0 / USD 0 |
| Complete actual cost / profit | Unknown / not established |

Logical work is an algorithm diagnostic, not money, total CPU, verification cost or customer value. Fictional accounting amounts are excluded from actual profit.

## Implementation and decision method

`flywheel-offering/v1` records a beneficiary, useful outcome, tangible deliverable, scope, exact acceptance criteria, internal need evidence, demand uncertainty, matched human-service reference, required capabilities, permitted tools, candidate prices, capacity, turnaround, complete-cost ceiling, risks and limits. Consecutive revisions reset qualification. They cannot inherit a previous version's success.

The separate stages are hypothesis, executed prototype, technical qualification, commercial assessment and authorized offering. A completed model response cannot advance an offering. Prototype qualification requires its own actual enterprise launch and exact terminal executor record. Technical qualification uses the existing independently signed, rights-bound general acceptance and complete A accounting. Its current implemented adapter remains finite service placement. Commercial forecasts remain signed assumptions. Their evidence is checked again when considering and dispatching production, including after authorization. A negative assessment stays recorded and cannot authorize production. Even a positive fictional assessment does not establish a live customer or grant permission to make external commitments.

The bounded decision policy considers production, assessment, Gym, specialist invocation, retention, sharing, prototyping, holding, deferring and stopping. It maximizes conservative forecast benefit minus the cash-cost cap and separately declared opportunity cost over the supplied feasible alternatives. Ties choose lower cash cost, then a stable identifier. This is a transparent finite policy, not a universal optimum or calibrated demand predictor. Alternative prices can compete in the same menu; no discount or reinvestment fraction is fixed.

Current journal evidence, source/configuration, current capability eligibility, reserves, unknown liabilities, occupied capacity, time and monetary bounds feed each decision. Offering reservations are scoped to their own offering while the global slot ceiling remains separate. Legacy unscoped reservations stay conservative until reconciliation. Unknown costs block additional funded work. The decision is advice until a separate signed exact permit authorizes the existing executor. Production and prototype missions must preserve the offering's beneficiary, purpose, deliverable, acceptance criteria, mode, tools and budgets. One decision permits one attempt. All alternative evidence and offering need records are rechecked in the dispatch transaction. An old decision refuses after relevant source, capability, control, offering, capacity or ledger changes.

`mission-run --enterprise-decision` connects the policy to the existing executor. Other lifecycle actions need their existing signed commands. Capability-dependent production composition remains unsupported; a requirement never invokes a capsule. Protected invocation retains the existing ENS/wallet and mission-permission boundaries.

## Executed comparisons and economics

The recorder executes one internal research-funded service-placement deliverable, checks it with the existing exhaustive oracle, records its offering qualifications, and runs the same origin into the existing Gym and protected capability lifecycle. Training searches retained executable workflows; proprietary model weights do not change. The frozen v2 is evaluated on 12 materially changed finite placement instances. Separate ordering, pruning, composition, strong conventional, private, shared and identical-content ordinary-library controls execute on four cases.

The technical path rejects admission: competent current frontier Beta and complete actual monetary costs are unmeasured. The controller selects holding funds and the adjacent disruption-aware rollout offering remains a hypothesis. Technical transfer is not a commercial qualification. The strong conventional procedure matches v2, identical-content access controls have the same algorithmic result, and composition adds coordination work. These observations do not demonstrate a graph advantage, commercially newly feasible work or sustained compounding.

A separate positive **fictional protocol episode** starts with its own connected offering/execution origin and invokes a pinned capsule through real signatures, encryption, loading and computation. Model, ENS and billing attestations are fictional. This tests the protected relay mechanism; it is not actual authenticated mainnet Agent B, real model work or net Alpha. The rejected technical episode is never relabeled as admitted.

Actual provider calls and customer collections are zero. Complete actual research cost and profit remain unknown. The illustrative equivalent human-service reference is 40 team hours at USD 100/hour (USD 4,000), disclosed as an estimate, not observed market pricing. USD 1,000/2,000/3,000 prices are hypotheses, not accepted quotations. Conformance seed capital and

category amounts are fictional. USDC operating economics, accounting USD units and applicable $AGIALPHA utility remain distinct.

Required development for the internal deliverable is charged once in A. Additional Gym development is C, and incremental assessment/retention/serving is B. No new controller fee is added to the ledger, and no paid-delivery work is billed again as research. Actual accounting still requires independent attribution of every resource category, corrections, failed attempts, refunds and upkeep. Opportunity cost is a decision input, not a second cash charge. Profitable Beta-only delivery is a valid policy without any retained capability.

## Failure boundaries and remaining qualification

Attempts reserve capacity before I/O and consume their decision permanently. Failed/interrupted executions retain uncertain liabilities and capacity. No automatic retry or restoration of authority occurs. To release a terminal failed attempt, first stop the enterprise, reconcile and seal all A liabilities with the existing accounting authority, then submit a signed `reconcile` command naming the exact launch head and current evidence. Terminal receipts must match the reserved permit, mission and profile. A lost launch finalization can be reconciled from that bound terminal failure; the consumed decision stays consumed. An attempt with no executor receipt or with an in-flight receipt requires separate operator investigation; this version deliberately cannot clear it automatically.

Directed regression tests exercise delivery, negative/positive fictional assessment, no-learning production, permission widening, stale evidence, roles, expiry, replay, version reset, unknown costs and interrupted recovery. The isolated full setup also reported dependency conflicts; a clean bootstrap is not established. Exact-source hosted review, required CI, full release coverage and installed/public qualification remain separate gates. Preserve the v1.46 main publication reservation; this child does not dispatch another publisher or replace existing papers.

Live qualification requires the existing minimum approved resources: model account with quota, liability caps, hard budget and billing; qualified sandbox/browser; protected HTTPS/storage/key/checkpoint custody; two approved mainnet RPC readers with checked code pins; two distinct owner-controlled eligible agent wallets; fresh rights-cleared missions, actual competent frontier comparisons and independent acceptance. No credentials, paid infrastructure, persistent access, financial transactions or external sales were provisioned. General capability-dependent composition also needs further engineering, not just credentials.

Official interfaces were checked on 27 October 2026: Astra, Terra, computer use, pricing. Current standard short-context token rates per million are Astra USD 10 input / 50 output and Terra USD 1 input / 5 output. Context, cache, service tier and tool charges require their applicable published terms. Astra supports low through max reasoning; Terra additionally supports none. Both roles use explicit approved profiles, and Terra tool calls require Responses. Published support proves no account entitlement or deployment qualification. ChatGPT Pro is not an API model identifier. This contribution changes no model default or credentials.

## Reproducible companion

The companion contains contracts, recorder, oracle, regressions, bilingual inspector and operator guide. Run `PYTHONPATH=. python scripts/record_enterprise_flywheel.py --output NEW.json` from a development checkout; evidence never overwrites a prior output. Build the paper using `python scripts/build_enterprise_paper.py`. Recorded runtime, recorder and fixture hashes must match first. Earlier packets retain their historical scope; model or tool changes require new qualification.

Evidence packet SHA-256:

`cd3cb8b766cdab9d39633f127f8894af03f5f082fe6ab34f2ef715de61521422`

Operator contracts are in `docs/agent/ENTERPRISE_NEXT_ADVANTAGE.md`.